

The divergence-free cofactor identity

MathlibAnnex.Piola.divergence_cofactorRowField_eq_zero

theorem

Cancels the Hessian terms that arise when differentiating a cofactor row.

Statement

Let $H : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be C^2 , and fix $i \in \{1, \dots, n\}$. Let e_j be the j th standard coordinate vector, with coordinates $(e_j)_q = \delta_{jq}$. Define the scalar coordinate maps and the derivative matrix by

$$H_k(x) := (H(x))_k, \quad A(x)_{kq} := \partial_q H_k(x).$$

Thus rows are output coordinates and columns are differentiation directions. Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the fixed i th cofactor row field:

$$F_j(x) := (F(x))_j = \det(A(x)[i \leftarrow e_j^\top]), \quad F(x) = C_i(A(x)).$$

Then, at every x ,

$$\operatorname{div} F(x) := \sum_{j=1}^n \partial_j F_j(x) = 0.$$

Assumptions

The hypothesis is twice continuous differentiability on the whole coordinate space. There is no invertibility, compact-support, or boundary assumption. A row i is supplied; the zero-dimensional case consequently has no row to check.

Conclusion

Every cofactor row field of the derivative is divergence-free, pointwise. The input and output coordinate indices remain distinct in the cancellation below.

Notes

Here H_k and F_j are scalar functions, whereas H and F are vector-valued maps. For any C^1 vector field V , $\partial_j V_j(x) = (DV(x)e_j)_j$. Thus taking the derivative in direction e_j and then taking coordinate j is precisely the j th term in the divergence.

Proof route

Differentiate one coordinate $F_j(x) = \det(A(x)[i \leftarrow e_j^\top])$ first. Rowwise differentiation and row multilinearity express it using $\partial_j \partial_q H_k$. In the divergence sum, the terms indexed by (j, q) and (q, j) cancel: their second derivatives agree and their two-row determinants have opposite signs.

▼ Proof steps

1. **Differentiate one cofactor coordinate, not the whole divergence at once.** Write the k th row of $A(x)$ as

$$r_k(x) = (\partial_1 H_k(x), \dots, \partial_n H_k(x)).$$

For fixed j ,

$$F_j(x) = \det(r_1(x), \dots, r_{i-1}(x), e_j^\top, r_{i+1}(x), \dots, r_n(x)).$$

Only the rows $r_k(x)$ with $k \neq i$ vary. The finite product rule in the determinant expansion therefore gives

$$\partial_j F_j(x) = \sum_{k \neq i} \det((A(x)[i \leftarrow e_j^\top])[k \leftarrow \partial_j r_k(x)]).$$

This is the rowwise derivative formula: each summand differentiates just one nonconstant row. The C^2 assumption makes every row being differentiated a C^1 map.

2. **Expand the differentiated row and then take the divergence.** The coordinate row satisfies

$$\partial_j r_k(x) = (\partial_j \partial_q H_k(x))_{q=1}^n = \sum_{q=1}^n \partial_j \partial_q H_k(x) e_q^\top.$$

Define, displaying both replaced rows,

$$B_{ikjq}(x) := (A(x)[i \leftarrow e_j^\top])[k \leftarrow e_q^\top].$$

Linearity of the determinant in row k turns Step 1 into

$$\partial_j F_j(x) = \sum_{k \neq i} \sum_{q=1}^n \partial_j \partial_q H_k(x) \det B_{ikjq}(x).$$

Finally sum over j and reorder these finite sums:

$$\operatorname{div} F(x) = \sum_{k \neq i} \sum_{j=1}^n \sum_{q=1}^n \partial_j \partial_q H_k(x) \det B_{ikjq}(x).$$

3. **Identify the two symmetries.** For each fixed $k \neq i$, equality of mixed second derivatives gives

$$\partial_j \partial_q H_k(x) = \partial_q \partial_j H_k(x).$$

The matrices $B_{ikjq}(x)$ and $B_{ikqj}(x)$ differ by interchanging rows i and k : these rows are $e_j^\top, e_q^\top$ in the first and $e_q^\top, e_j^\top$ in the second. Hence

$$\det B_{ikjq}(x) = -\det B_{ikqj}(x).$$

For $j = q$, the two rows coincide and each determinant is zero. There is no matrix-invertibility hypothesis in either step.

4. Perform the cancellation. For a fixed $k \neq i$, call the inner finite sum $S_k(x)$. Relabel j and q , and then use Step 3:

$$\begin{aligned} S_k(x) &:= \sum_{j,q} \partial_j \partial_q H_k(x) \det B_{ikjq}(x) \\ &= \sum_{j,q} \partial_q \partial_j H_k(x) \det B_{ikqj}(x) \\ &= - \sum_{j,q} \partial_j \partial_q H_k(x) \det B_{ikjq}(x) = -S_k(x). \end{aligned}$$

Thus $2S_k(x) = 0$ over $\mathbb{R}$, so $S_k(x) = 0$. Substitution into Step 2 gives $\operatorname{div} F(x) = \sum_{k \neq i} S_k(x) = 0$.

Main citations

- Exact declaration and its source — `MathlibAnnex.Piola.divergence_cofactorRowField_eq_zero`
- Row-replacement cofactor convention — `MathlibAnnex.Piola.cofactorRow`
- Standard matrix entries (private source helper) — `MathlibAnnex.Piola.stdMatrix`
- Two designated row replacements — `MathlibAnnex.Piola.twoRowReplacement`
- Cofactor row field of a derivative — `MathlibAnnex.Piola.cofactorRowField`
- Antisymmetry of two row replacements — `MathlibAnnex.Piola.det_twoRowReplacement_swap`
- Symmetric–antisymmetric finite-sum cancellation —
`MathlibAnnex.Piola.sum_symmetric_mul_antisymmetric_eq_zero`
- Hessian contraction with two replacement rows —
`MathlibAnnex.Piola.sum_hessian_twoRowReplacement_eq_zero`
- Hessian output coordinate and input directions — `MathlibAnnex.Piola.hessianCoordinate`
- Symmetry of the two Hessian directions — `MathlibAnnex.Piola.hessianCoordinate_comm`
- Cancellation for the derivative Hessian —
`MathlibAnnex.Piola.sum_hessianCoordinate_twoRowReplacement_eq_zero`
- Piola identity reduced to its rowwise expansion —
`MathlibAnnex.Piola.piola_divergence_eq_zero_of_expansion`

Lean source signature (exact)

```
theorem divergence_cofactorRowField_eq_zero
  {n : ℕ} {H : (Fin n → ℝ) → (Fin n → ℝ)}
  (hH : ContDiff ℝ 2 H) (i : Fin n) (x : (Fin n → ℝ)) :
  ∑ j, fderiv ℝ (cofactorRowField H i) x (Pi.single j 1) j = 0
```

In the source

```
{n : ℕ} {H : (Fin n → ℝ) → (Fin n → ℝ)} (hH :
ContDiff ℝ 2
H)
```

Mathematical meaning

The map $H : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is C^2 on the whole space.

In the source	Mathematical meaning
<code>(i : Fin n) (x : (Fin n → ℝ))</code>	The fixed cofactor row i and arbitrary evaluation point x .
<code>cofactorRowField H i</code>	The vector field $F(x) = C_i(DH(x))$, with derivative matrix entries $A(x)_{kq} = \partial_q H_k(x)$: output index k , input direction q .
<code>Pi.single j 1</code>	The standard direction e_j , with coordinates $(e_j)_q = \delta_{jq}$.

In the source	Mathematical meaning
<code>fderiv ℝ (cofactorRowField H i) x (Pi.single j 1)</code> <code>j</code>	Differentiate the vector field F in direction e_j , then take its j th output coordinate: $(DF(x)e_j)_j = \partial_j F_j(x)$.
<code>Σ j, fderiv ℝ (cofactorRowField H i) x (Pi.single j 1) j = 0</code>	The conclusion is the pointwise identity $\operatorname{div} F(x) = \sum_j \partial_j F_j(x) = 0$.

▼ Exact surrounding binder context (separate excerpt)

```

noncomputable section

open Set
open scoped BigOperators Topology

namespace MathlibAnnex
namespace Piola

```

▼ Exact content identity

Declaration: `MathlibAnnex.Piola.divergence_cofactorRowField_eq_zero`
Accepted content SHA-256: `e586a6536e27c03ab4425c4b10c551655da32a43be1170c424c4b06bb2f20a4d`
Accepted source guide SHA-256: `cca27385be48b2092fc739c16a4889a53b2d3d714458b0ba9aa893468bfe29d2`
Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`